Korea Institute of Science and Technology

Surface analysis:

Photoelectron spectroscopy

Introduction and Principles of

Surface Analysis Techniques

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Overview of this lecture

목 차

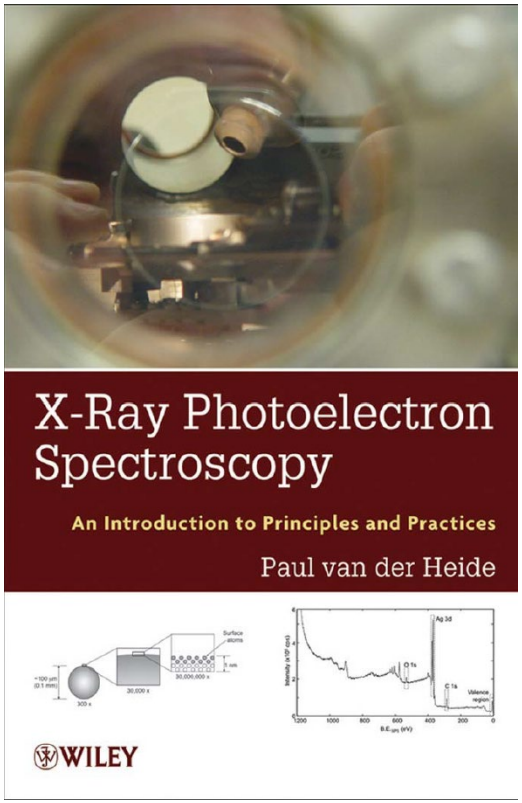
1 Textbook and Reference Materials

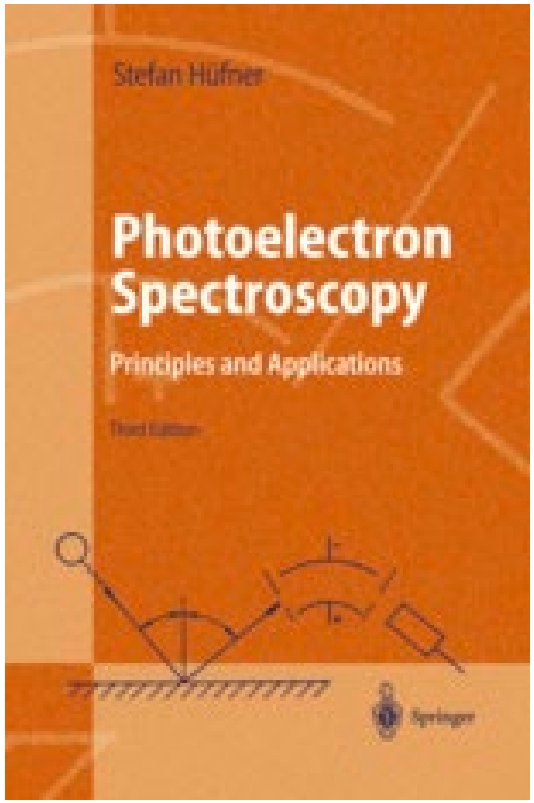
2 What Are Surfaces and Interfaces?

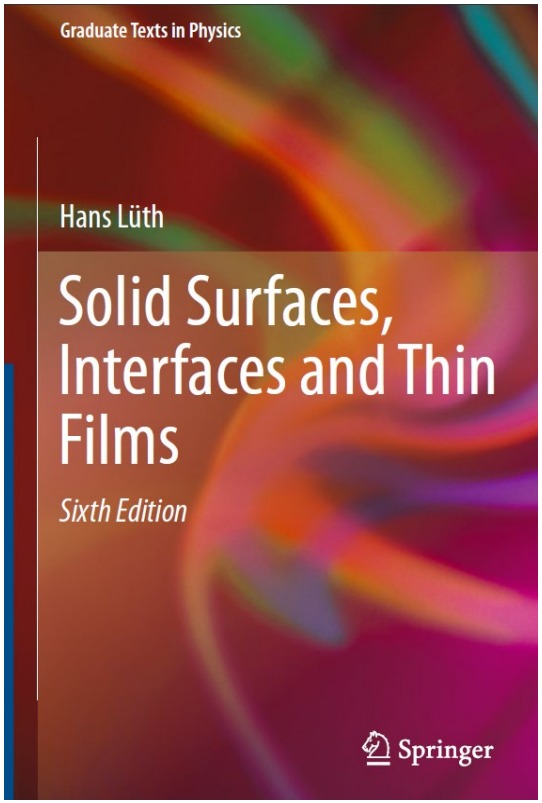
3 The Importance of Surfaces and Interfaces

4 Resolution and Detection Limits

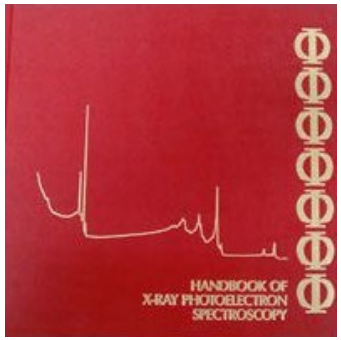
Textbook and Reference Materials

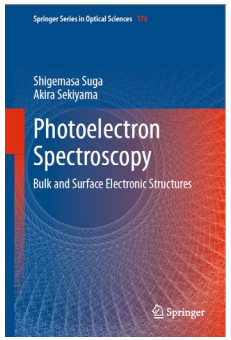






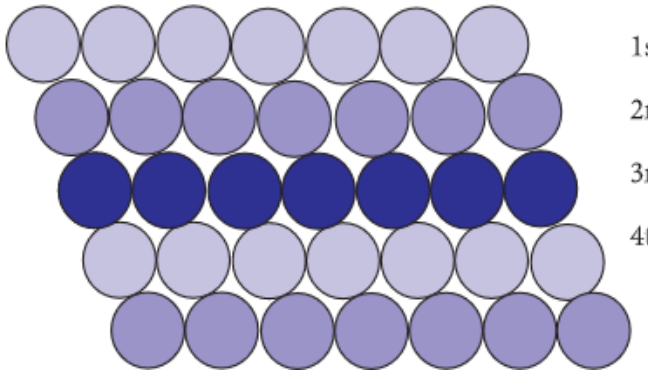
1. Introduction
2. Atoms, Ions, and Their Electronic Structure
3. XPS Instrumentation
4. Data Collection and Quantification
5. Spectral Interpretation
6. Some Case Studies





What Are Surfaces and Interfaces?

Definition :

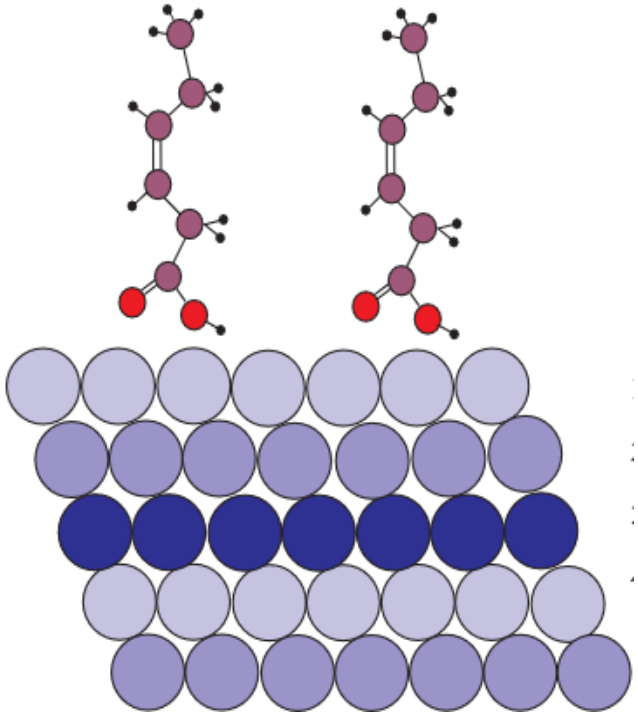


1st layer (surface)

2nd layer (interface)

3rd layer (medium)

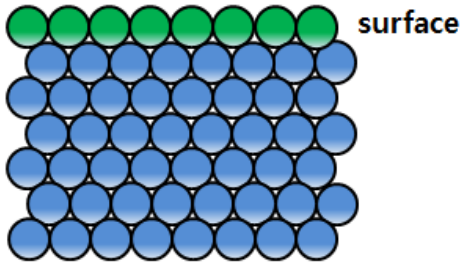
4th and 5th layers (bulk)

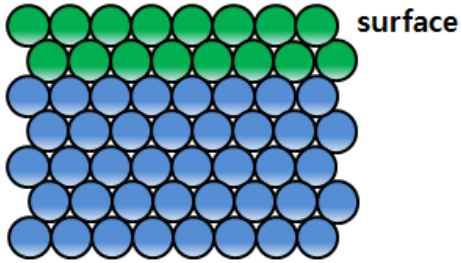


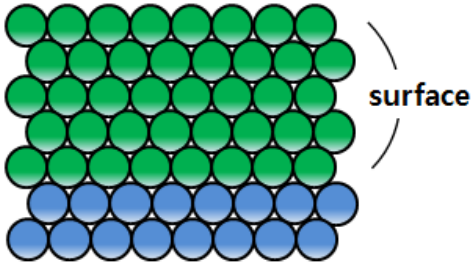
What Are Surfaces and Interfaces?

How can we define the Surface or interface?

" A surface is the **region** that dictates how the solid or liquid **interacts with its surroundings**"





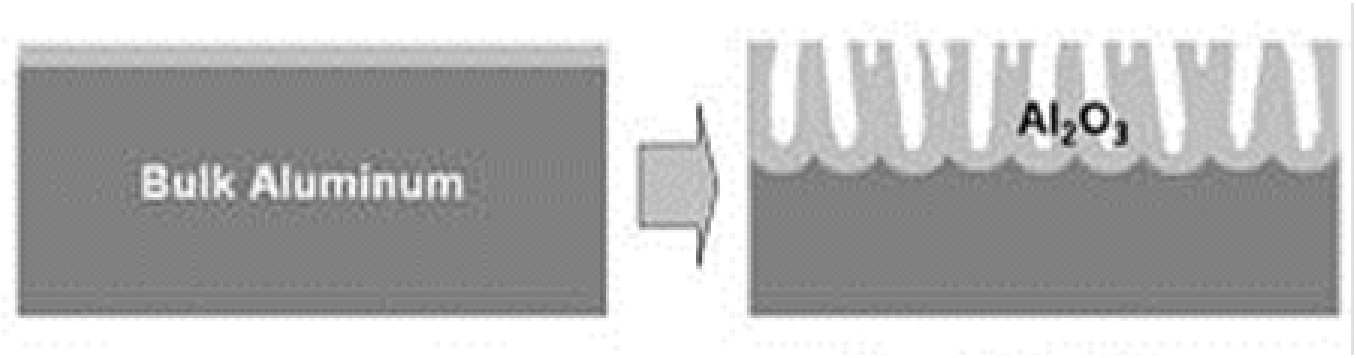


The importance of Surfaces and Interfaces

Surface is crucial
for understanding material/properties!"

The importance of Surfaces and Interfaces

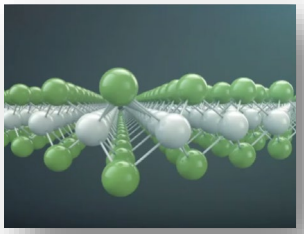
"Almost all materials have different properties at their surface and in their bulk



For example, in the case of aluminum (Al), it exhibits strong reactivity, but the oxidized surface (Al2O3) acts as a passivating layer, making it appear as if it has low reactivity.

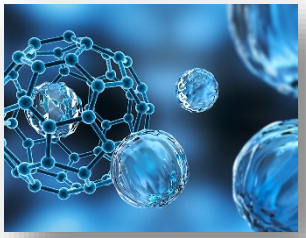
The importance of Surfaces and Interfaces

Semiconductor



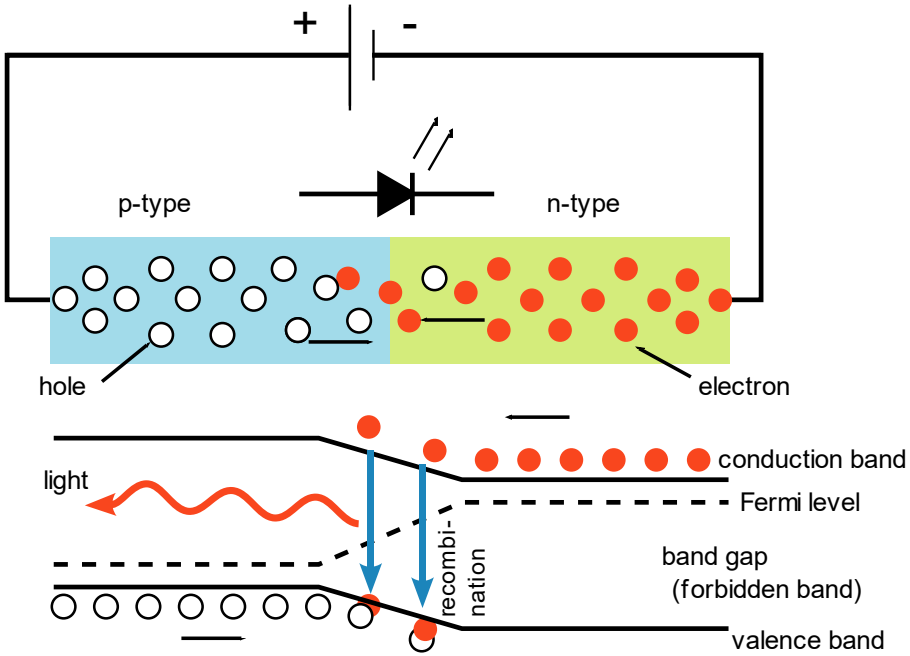
OLED

catalyst



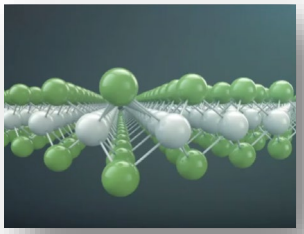
Battery

"Schematic Diagram of a P-N Diode Semiconductor Structure

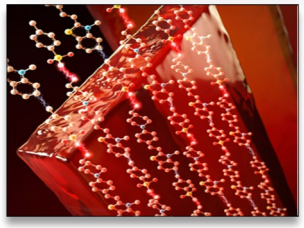


The importance of Surfaces and Interfaces

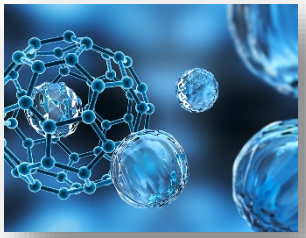
Semiconductor



OLED

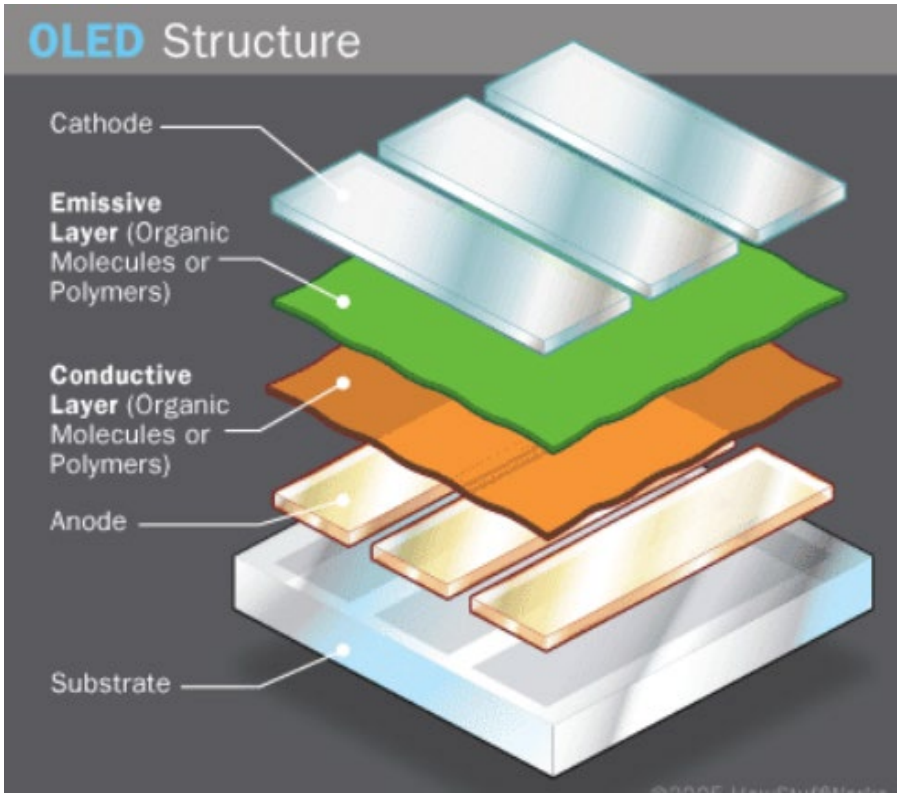


catalyst



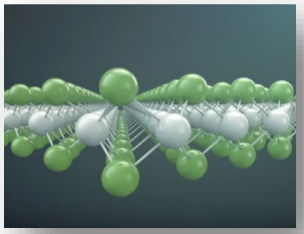
Battery

Schematic Diagram of an Organic Light-Emitting Diode



The importance of Surfaces and Interfaces

Semiconductor



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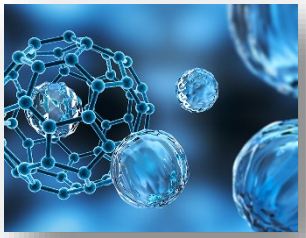
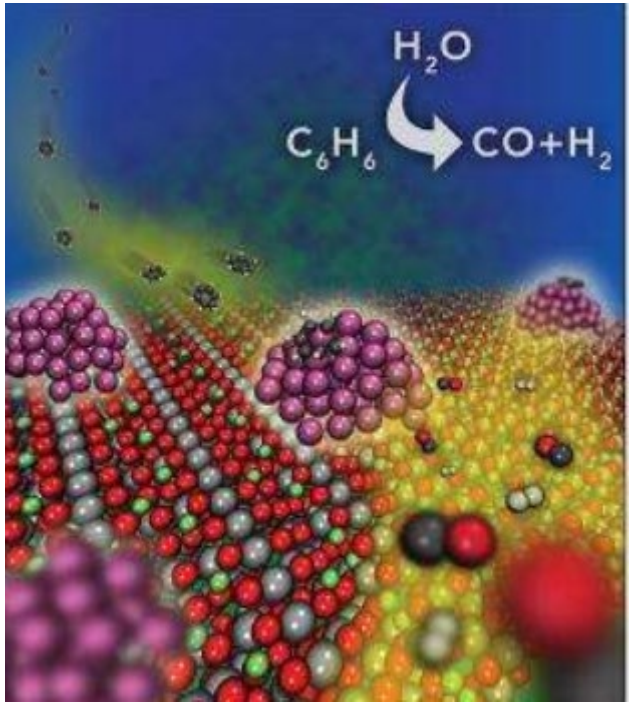


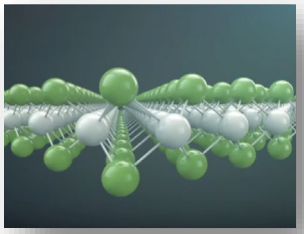
Battery

Schematic Diagram of Methane -> CO + H2 Catalytic Reaction



The importance of Surfaces and Interfaces

Semiconductor



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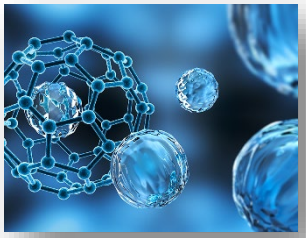
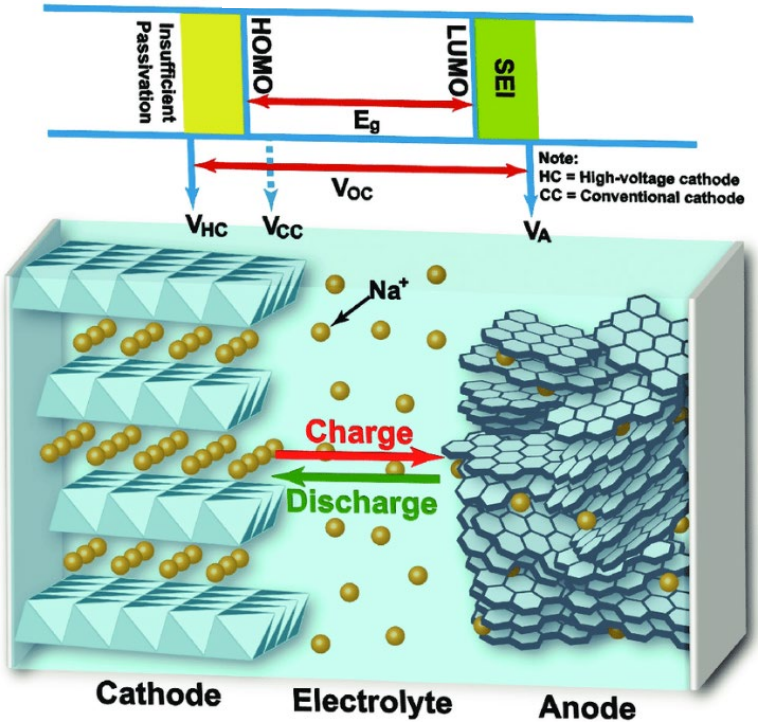
catalyst



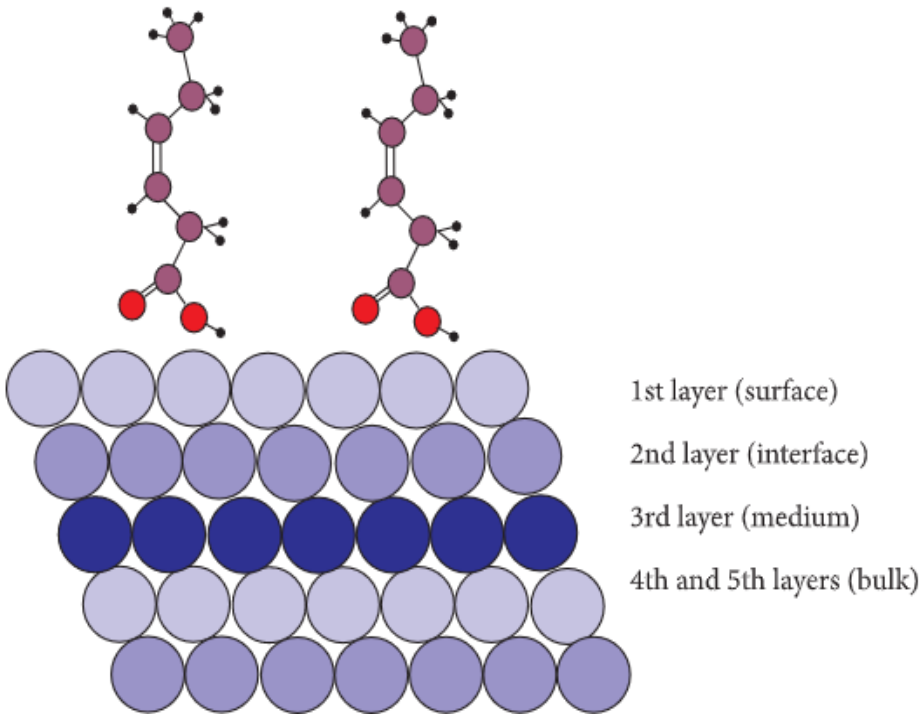
Battery

Schematic Diagram of a Li-ion Battery Structure



https://doi.org/10.1002/aenm.201701785

Resolution and Detection Limits

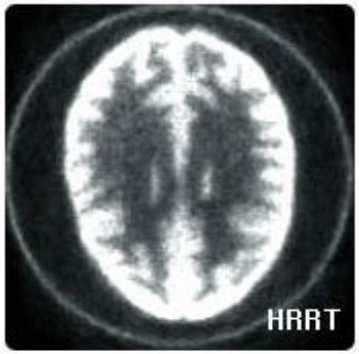


What will we measure?

Resolution and Detection Limits

Spatial Resolution:

Detection Limits:



Resolution and Detection Limits

Spatial Resolution:

Detection Limits:

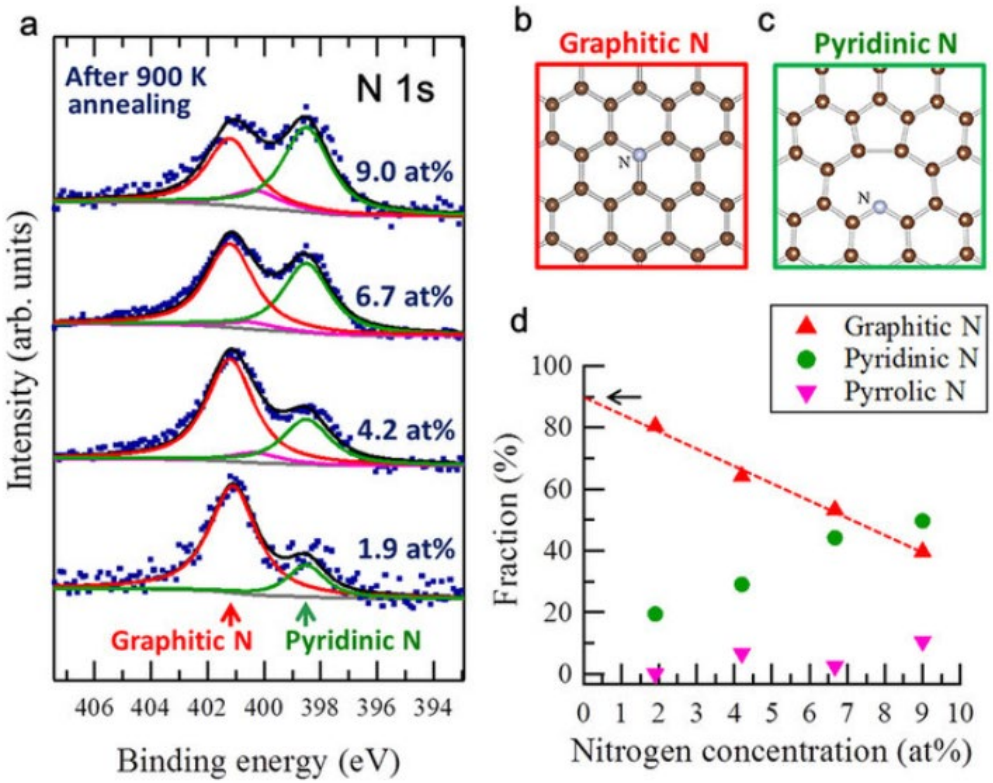
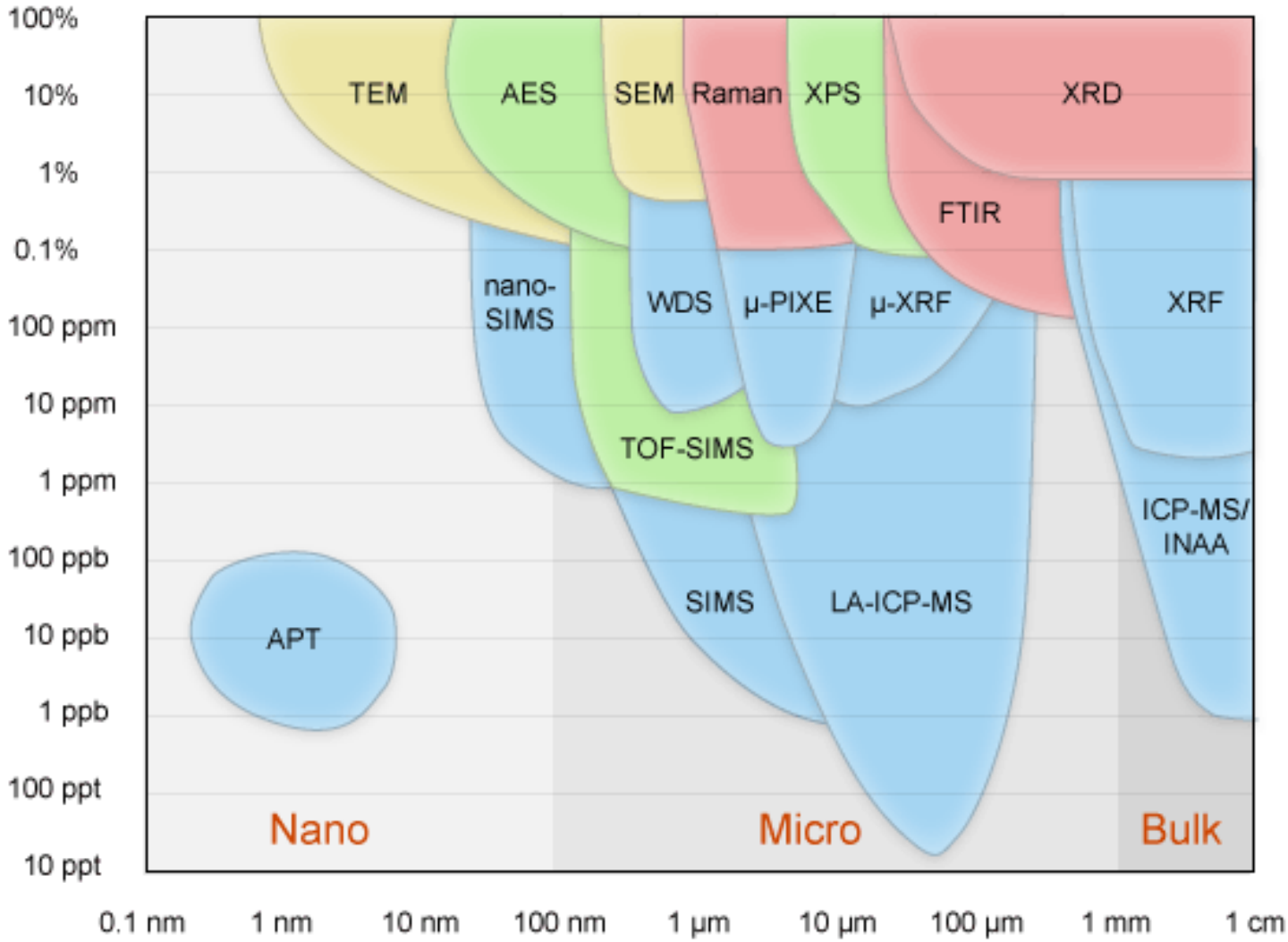


Figure 2: XPS spectra of nitrogen-doped graphite surfaces showing graphitic-N dominance at low N concentration.: (a) XPS N1s core level spectra of nitrogen-doped graphite surfaces with different N concentrations. (b,c) Schematic images of graphitic N and pyridinic N. (d) The fractions of graphitic N, pyloric N and pyridinic N in nitrogen as a function of nitrogen concentration of the nitrogen-doped graphite surface.

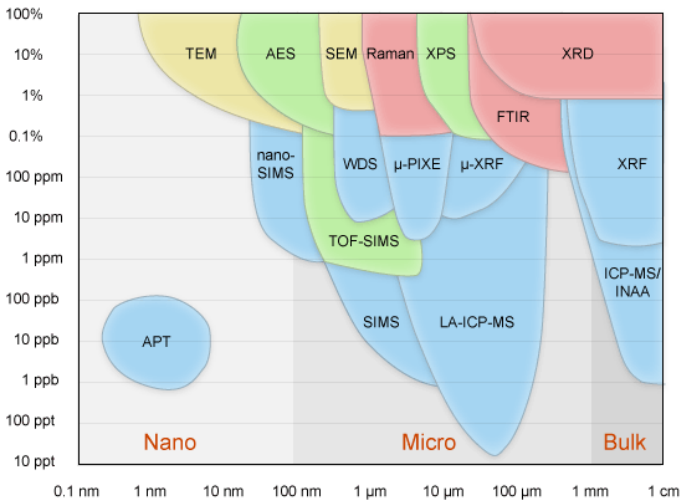
DOI: 10.1038/srep16412

Resolution and Detection Limits



https://serc.carleton.edu/

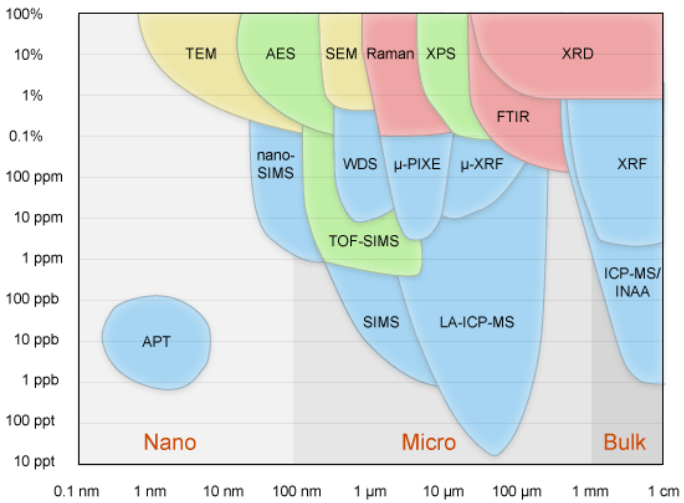
Resolution and Detection Limits



Principles of Analytical Instruments

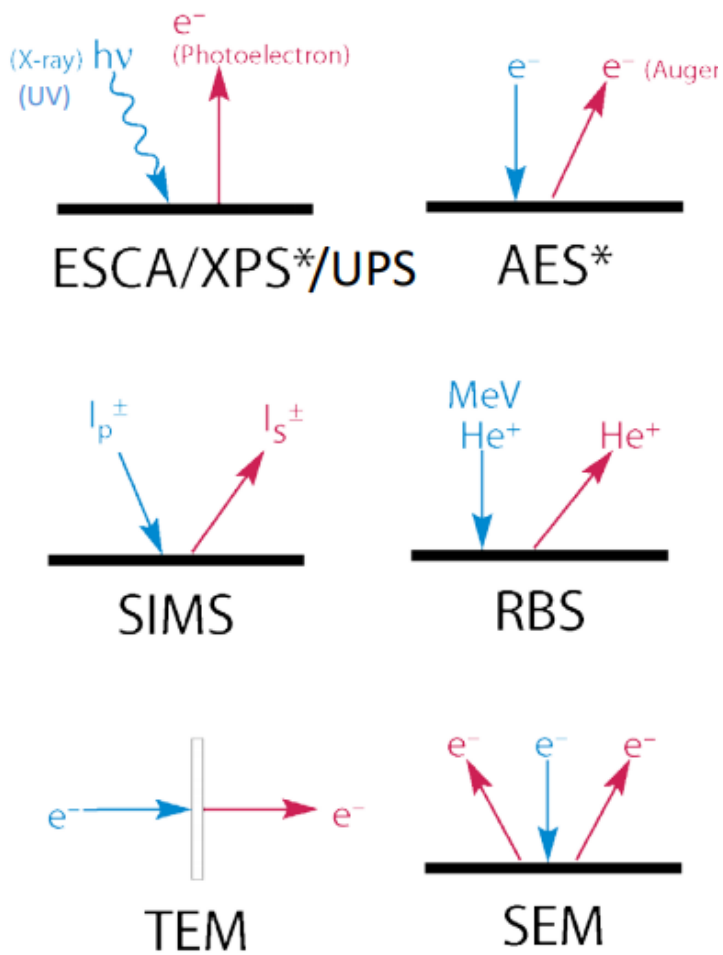
In \ Out	Light	Electron	Particle
Light	XRD, XRF Raman, FTIR	XPS UPS	-
Electron	IPES	AES SEM TEM	-
Particle	-	-	TOF-SIMS SIMS ISS

Resolution and Detection Limits

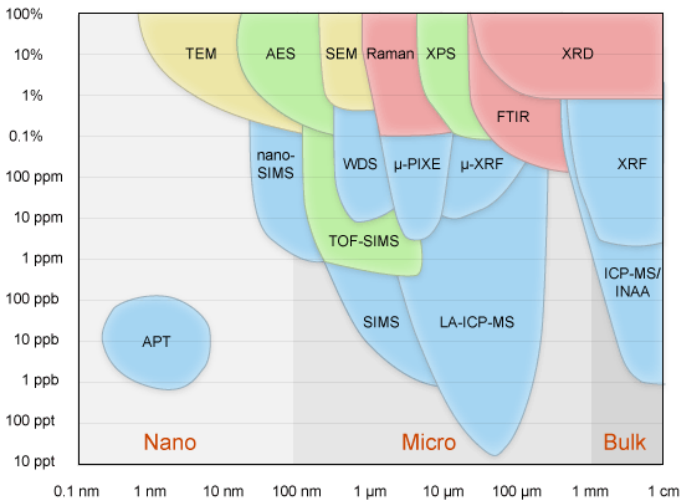


In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS

Principles of Analytical Instruments



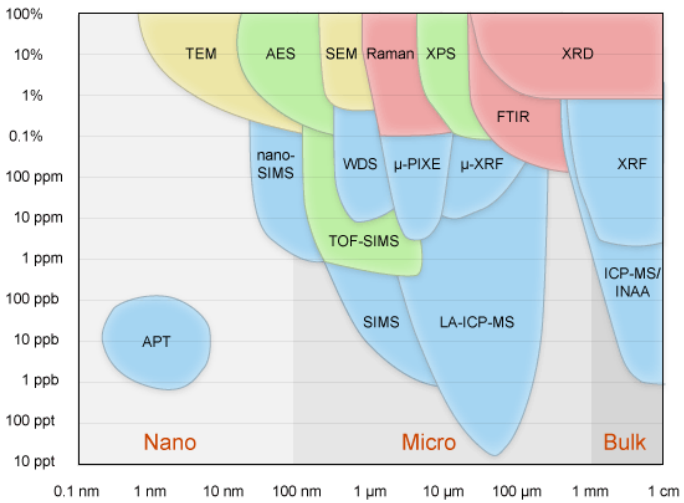
Resolution and Detection Limits



Why is there a difference in spatial resolution

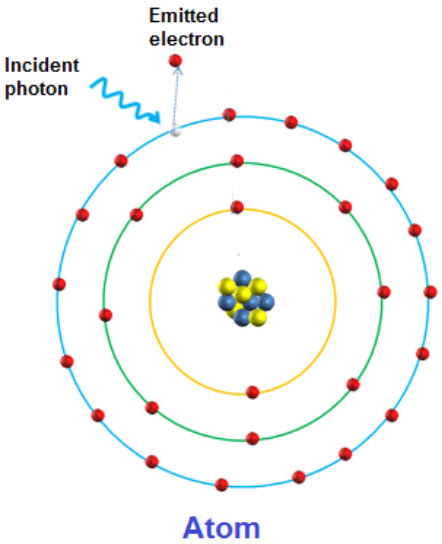
In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS

Resolution and Detection Limits

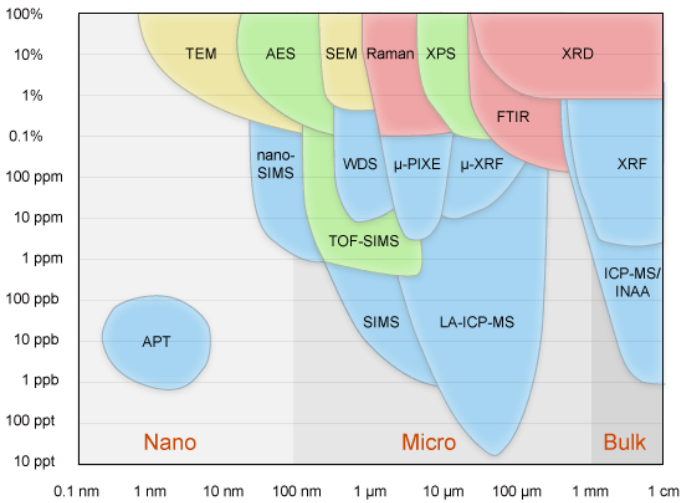


XPS

In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS

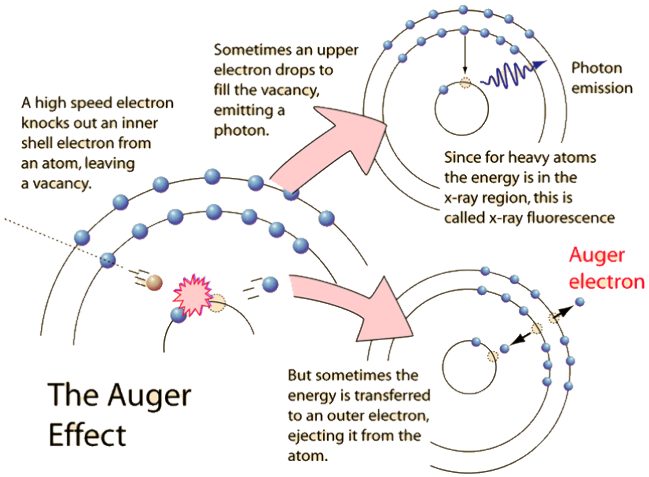


Resolution and Detection Limits

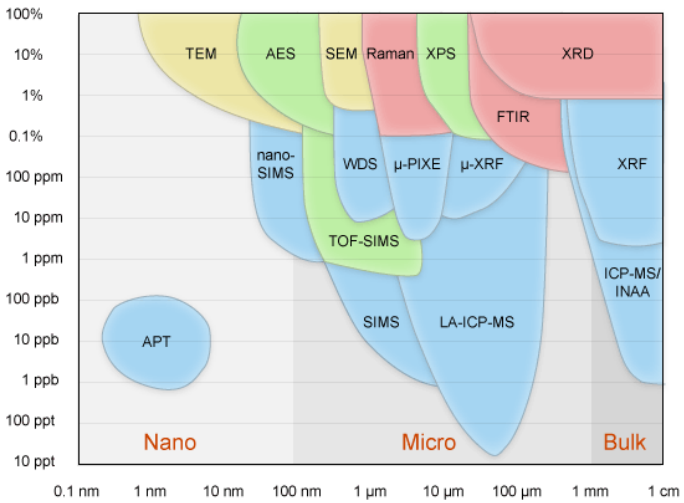


AES

In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS

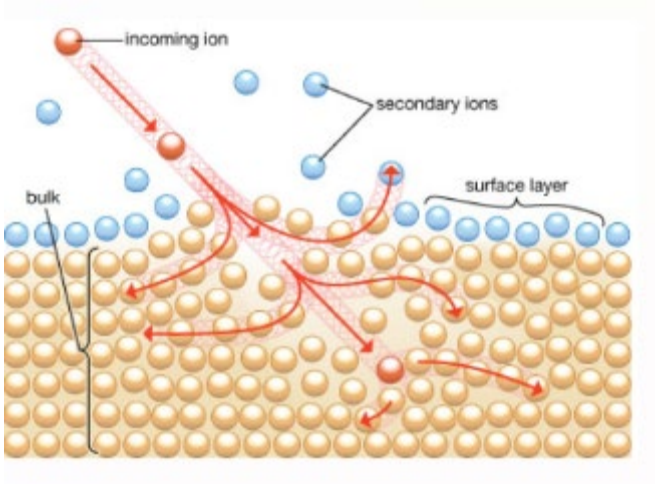


Resolution and Detection Limits

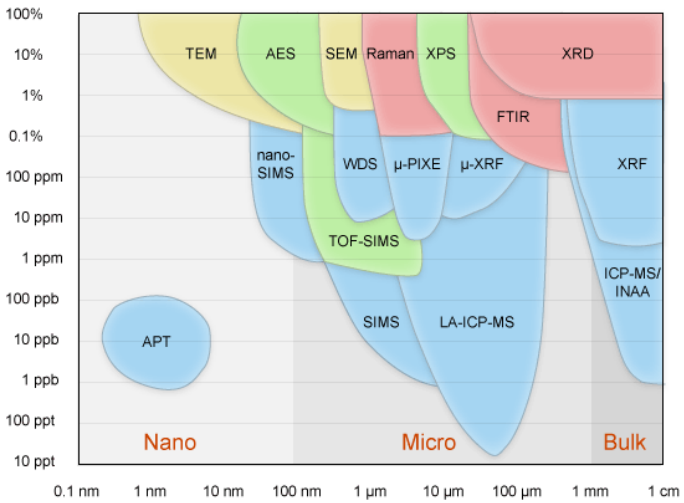


SIMS

In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS

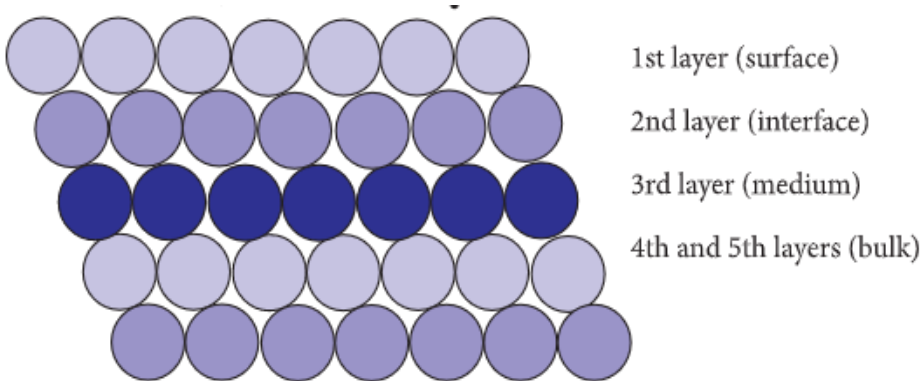


Resolution and Detection Limits

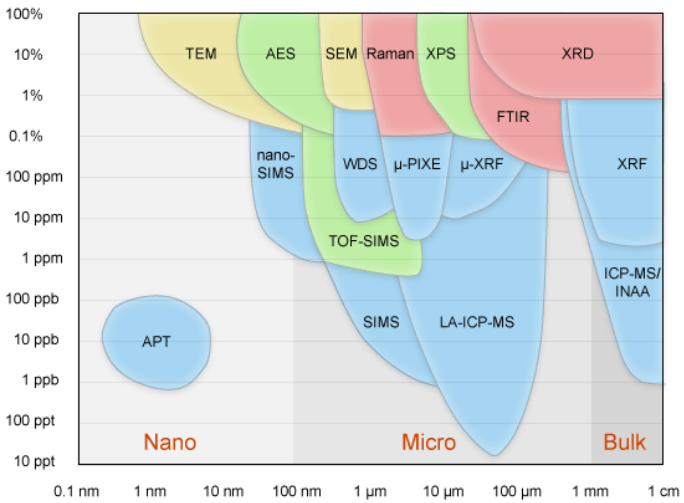


Depth Resolution

In \ Out	빛	전자	입자
빛	XRD, XRF Raman, FTIR	XPS UPS	-
전자	IPES	AES SEM TEM	-
입자	-	-	TOF-SIMS SIMS ISS



Resolution and Detection Limits



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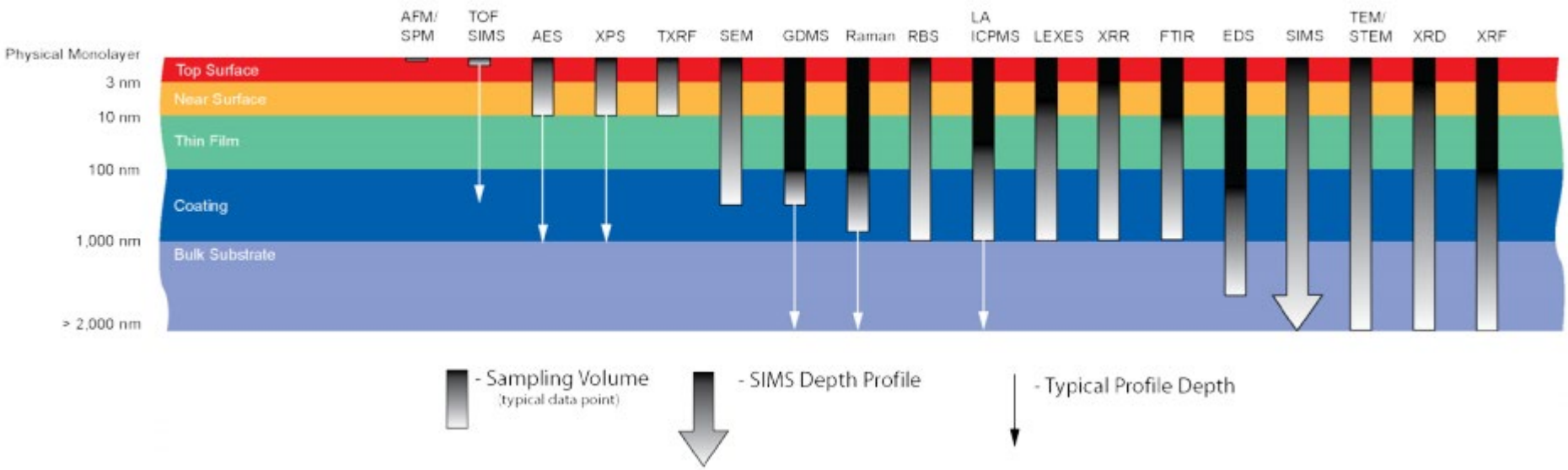

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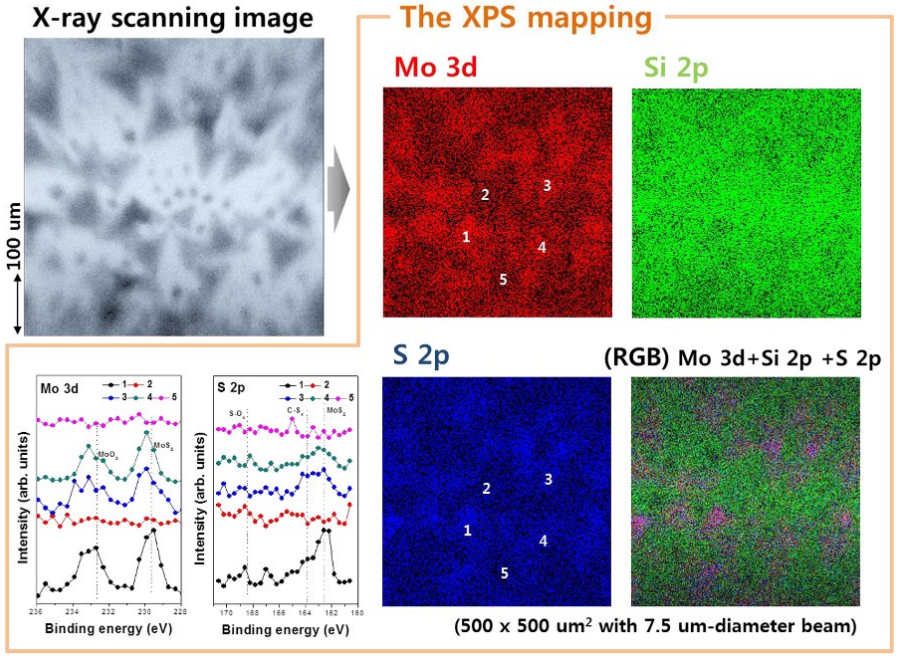
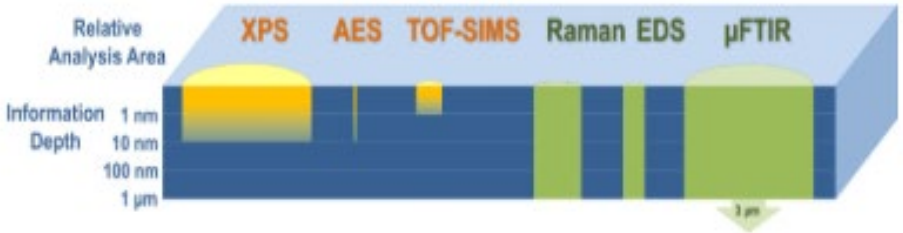
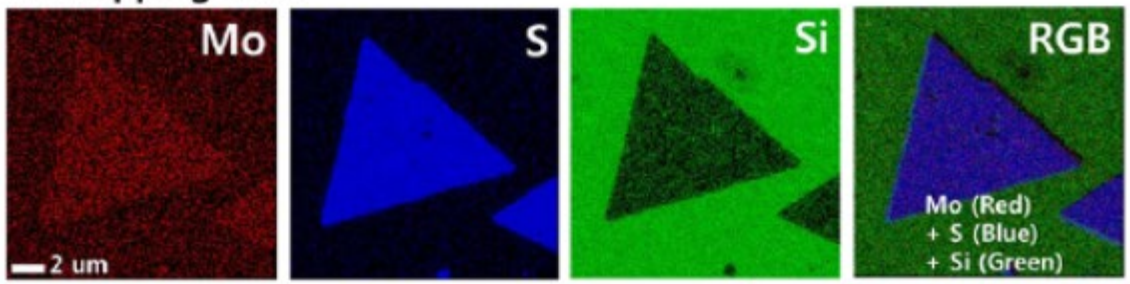
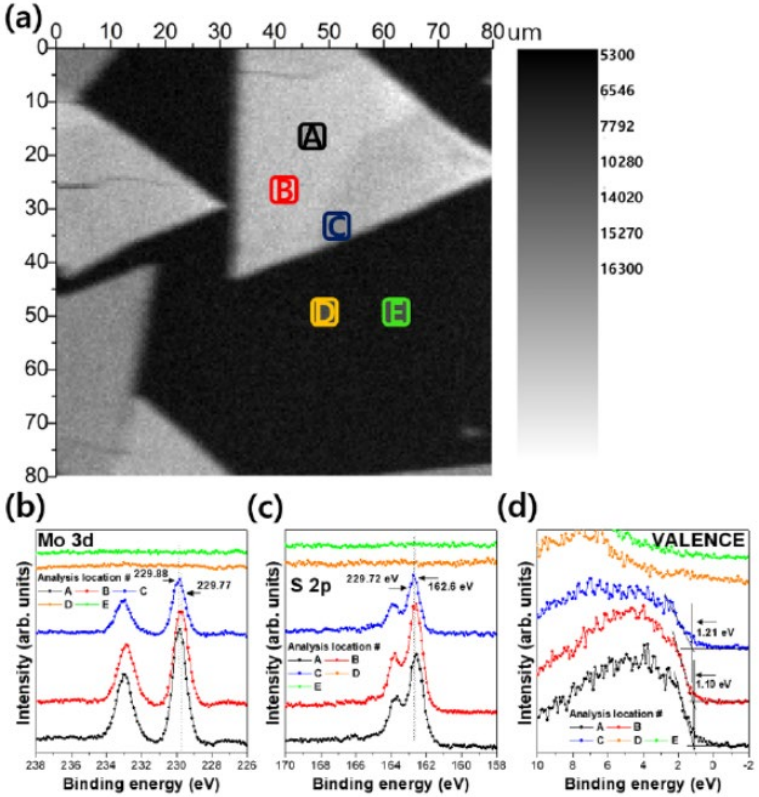


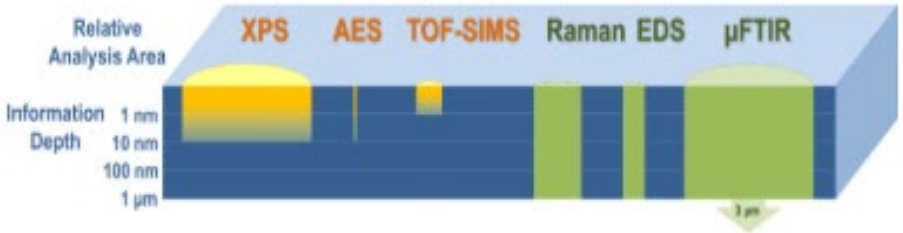


(f) AES mapping data

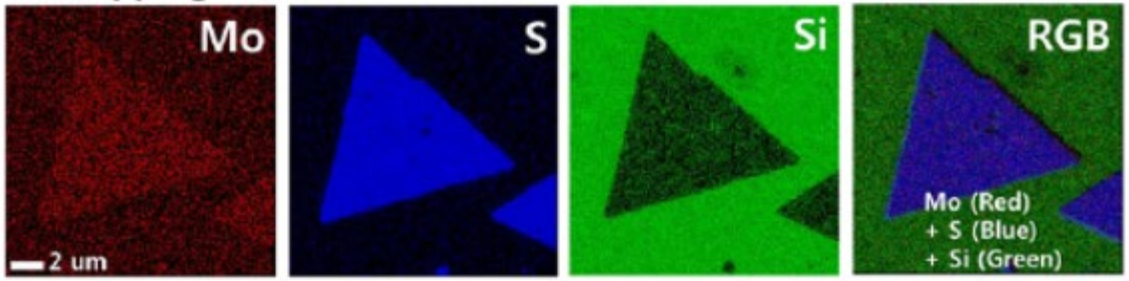


Resolution and Detection Limits





(f) AES mapping data



Thanks for listening.